

Homework 2

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Problem 1. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions.

1. Show that if $g \circ f$ is injective, then f is injective
2. Show that if $g \circ f$ is surjective, then g is surjective.

Claim 1a. Show that if $g \circ f$ is injective, then f is injective

Proof. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. Suppose $g \circ f$ is injective. Assume the contrary, f is not injective. Then by the negation of the definition of injectivity, $\exists x, y \in A$, where $x \neq y$ but $f(x) = f(y)$. Then $g(f(x)) = (g \circ f)(x) = (g \circ f)(y) = g(f(y))$. By definition of injectivity, since $(g \circ f)(x) = (g \circ f)(y) \implies x = y$. This leads to a contradiction because $x = y$ and $x \neq y$. Thus by contradiction, f must be injective. $\square$

Claim 1b. Show that if $g \circ f$ is surjective, then g is surjective

Proof. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. Suppose $g \circ f$ is surjective. Assume the contrary, g is not surjective. Then by the negation of the definition of surjectivity, $\exists y \in C \ni \nexists b \in B \ni g(b) = y$. Thus $\nexists a \in A \ni f(a) = b$, thus $\nexists a \in A \ni (g \circ f)(a) = g(f(a)) = g(b) = y$. However by definition of surjectivity, $\exists a \in A \ni (g \circ f)(a) = y$. Thus there is a contradiction, because $\exists a \in A \ni (g \circ f)(a) = y$ and $\nexists a \in A \ni (g \circ f)(a) = y$. Thus by contradiction, g is surjective. $\square$

Claim 2a. $\forall n \in \mathbb{N}$:

$$\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{1}{1(2)} + \frac{1}{2(3)} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

Proof. (Base Case): Let $n = 1$, $\sum_{i=1}^1 \frac{1}{i(i+1)} = \frac{1}{1(2)} = \frac{1}{2}$. $n + 1 = 2$. So $\sum_{i=1}^1 \frac{1}{i(i+1)} = \frac{n}{n+1}$.

Inductive Hypothesis: Assume $\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1}$ is true.

Inductive Step: $\sum_{i=1}^{n+1} \frac{1}{i(i+1)} = \sum_{i=1}^n \frac{1}{i(i+1)} + \frac{1}{(n+1)(n+2)}$. By Inductive Hypothesis,

$$\begin{aligned}
\sum_{i=1}^n \frac{1}{i(i+1)} + \frac{1}{(n+1)(n+2)} &= \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} \\
&= \frac{n(n+2) + 1}{(n+1)(n+2)} \\
&= \frac{n^2 + 2n + 1}{(n+1)(n+2)} \\
&= \frac{(n+1)^2}{(n+1)(n+2)} \\
&= \frac{n+1}{n+2}.
\end{aligned}$$

Thus, $\sum_{i=1}^{n+1} \frac{1}{i(i+1)} = \frac{n+1}{n+2}$. Let $k = n+1$. Thus, $\sum_{i=1}^k \frac{1}{i(i+1)} = \frac{k}{k+1}$.

Thus by induction, $\forall n \in \mathbb{N}$, $\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1}$ □

Claim 2b. $\forall n \in \mathbb{N}$,

$$n < 2^n$$

Proof. (Base Case): Let $n = 1$, $2^n = 2$ and $1 < 2$ so $n < 2^n$.

Inductive Hypothesis: Suppose $n < 2^n$

Inductive Step: By the inductive hypothesis, $n < 2^n \implies 2n < 2^{n+1}$. Since $n \in \mathbb{N}$, $n \geq 1$, $n+1 \leq n+n = 2n$. Thus $n+1 \leq 2n < 2^{n+1} \implies n+1 < 2^{n+1}$.

Thus by induction, $\forall n \in \mathbb{N}$, $n \leq 2^n$. □

Problem 3. Assume that T_1 and T_2 are two sets.

1. Assume that T_1 is finite, and that there exists a bijection from T_2 to T_1 . Prove that T_2 is also finite.
2. Same question replacing finite by denumerable

Claim 3a. Assume that T_1 is finite, and that there exists a bijection from T_2 to T_1 . Prove that T_2 is also finite.

Proof. Assume that T_1 is finite, and that there exists a bijection g from T_2 to T_1 . By the Uniqueness Theorem, $\exists! n \in \mathbb{N} \ni |T_1| = n$. By definition of finite, $\exists f : \mathbb{N}_n \rightarrow T_1$ where f is a bijection. Since the inverse of a bijection is a bijection, g^{-1} is a bijection, thus there is a bijection between T_1 and T_2 . The composition of 2 bijections is also a bijection, so $g^{-1} \circ f$ is a bijection from $\mathbb{N}_n \rightarrow T_2$. By definition of finite sets, T_2 is a finite set. □

Claim 3b. Assume that T_1 is denumerable, and that there exists a bijection from T_2 to T_1 . Prove that T_2 is also denumerable.

Proof. Assume that T_1 is denumerable, and that there exists a bijection from T_2 to T_1 . By definition of denumerable, $\exists f : \mathbb{N} \rightarrow T_1$ where f is a bijection. Since the inverse of a bijection is a bijection, g^{-1} is a bijection, thus there is a bijection between T_1 and T_2 . The composition of 2 bijections is also a bijection, so $g^{-1} \circ f$ is a bijection from $\mathbb{N} \rightarrow T_2$. By definition of denumerable sets, T_2 is a finite set. $\square$

Problem 4. *Is it possible to have a bijection from $\mathbb{N}$ to a proper subset of $\mathbb{N}$? Either prove that it is not possible, or give an example.*

Proof. $S := \{2n : n \in \mathbb{N}\}$ or the set of even numbers. $S \subset \mathbb{N}$. $f : \mathbb{N} \rightarrow S$, where $\forall n \in \mathbb{N}, f(n) = 2n$. $\forall x, y \in \mathbb{N} \ni f(x) = f(y) \implies 2x = 2y \implies \frac{1}{2}(2x) = \frac{1}{2}(2y) \implies x = y$. Thus by definition of injective, f is injective.

$\forall y \in S$, by definition of S , y is even. Thus $\frac{y}{2} \in \mathbb{N}$ and $f(\frac{y}{2}) = \frac{2y}{2} = y$. Thus f is surjective.

Thus f is a bijection. $\square$

Claim 5. *Let S be a set. Denote $P(S)$ the collection of all subsets of S . For instance if $S = \{1, 2\}$*

$$P(S) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

Show by induction that if S has n elements, then $P(S)$ has 2^n elements.

Proof. (Base Case): Let $n = 0$, $S = \emptyset$. Assume the contrary, $\exists A \in P(S) \ni \exists x \in A$, since $A \subset S$. By definition of subset, $x \in S \implies x \in \emptyset$ which is a contradiction. Thus $\nexists x \in A \implies A = \emptyset$. Thus the only subset of S is $\emptyset$. Thus $P(S) = \{\emptyset\}$. $|P(S)| = 1 = 2^0$.

Induction Hypothesis: Assume for all sets S with n elements, then $P(S)$ has 2^n elements.

Inductive Step: Let S be a set with $n + 1$ elements. Take a subset of S , A with n elements. $\exists! x \in S \ni x \notin A$. Thus $S = A \cup \{x\}$. For any subset $B \subset S$ or in other words, $B \subset P(S)$, $B \subset A$ or the union of a subset of A and $\{x\}$. By the induction hypothesis, there are 2^n subsets $B \in P(A)$, hence there are 2^n subsets of A . Hence there are 2^n subsets of the form $B \cap \{x\}$. Hence S has 2^n subsets which don't contain x and 2^n subsets which contain x . Thus $|P(S)| = 2^n + 2^n = 2(2^n) = 2^{n+1}$

Thus by induction, if S has n elements, then $P(S)$ has 2^n elements. $\square$